

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** of experience building and operating large-scale distributed systems at Google, YouTube, Microsoft, and Whatnot. Owned end-to-end infrastructure for platforms serving billions of requests daily, led teams delivering high-throughput services with on-call ownership. Technical expertise spans **C#, .NET, Python, SQL Server, Databricks**, and **Azure DevOps**. Drove latency reductions exceeding 40% across multiple systems and managed cross-functional partnerships to align technical roadmaps with business objectives. Works on scalable architectures and AI-driven solutions for high-growth environments. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages & Frameworks: C#, .NET, Python, Go, Java, TypeScript, JavaScript, HTML, CSS
Backend & Services: microservices, RESTful APIs, distributed systems, large-scale infrastructure
Data Engineering: SQL Server, Databricks, data pipelines, ETL processes
Platform & DevOps: Azure DevOps, CI/CD pipelines, Kubernetes, cloud infrastructure, on-call operations

Work Experience

Staff Software Engineer

YouTube | 02/2025 - Present

- Architected and deployed video processing pipelines handling millions of uploads daily, implementing fault-tolerant workflows that reduced processing failures by 25%.
- Designed scalable backend services using distributed systems principles, supporting billions of daily API calls with sub-100ms p99 latency targets.
- Led cross-functional initiatives with product and infrastructure teams to define technical roadmaps for new platform capabilities.
- Established monitoring and alerting frameworks for critical production systems, improving incident detection and resolution times.
- Mentored junior engineers on system design patterns and operational best practices across the organization.

Staff Software Engineer

Whatnot | 05/2024 - 02/2025

- Built real-time auction platforms supporting thousands of concurrent users, implementing event-driven architectures that handled 10x traffic spikes during peak sales events.
- Developed core backend services in **C#** and **.NET** for marketplace transactions, processing payments with high reliability and consistency.
- Established **CI/CD pipelines** with **Azure DevOps**, automating deployments and reducing release cycles from weekly to daily.
- Designed data pipelines using **SQL Server** and **Databricks** for analytics, enabling near-real-time dashboards for business stakeholders.
- Reduced database query times by 35% through query optimization and indexing strategies on high-traffic tables.

Staff Software Engineer/Manager

Google | 03/2015 - 04/2024

- Managed a team of 8 engineers building and maintaining search infrastructure serving billions of queries daily with strict availability requirements.
- Owned on-call rotations for critical services, maintaining 99.99% uptime through proactive monitoring and rapid incident response.
- Collaborated with partners in infrastructure and data teams to deliver platform-wide improvements affecting multiple product areas.
- Conducted technical interviews and hiring for senior engineering roles, building a high-performing team over 9 years.
- Migrated monolithic services to **microservices** architecture, improving deployment independence and system resilience across the stack.

Software Engineer

Google | 02/2011 - 02/2015

- Optimized data processing jobs reducing computational costs by 20% through algorithm improvements and resource scheduling.
- Built automated testing frameworks that increased code coverage and caught regressions before production deployment.
- Developed backend systems for search ranking infrastructure using **Python** and distributed processing frameworks.
- Participated in cross-team projects to enhance serving infrastructure for improved query response times.

Software Design Engineer II

Microsoft | 10/2007 - 12/2011

- Developed data access layers with **SQL Server**, optimizing stored procedures for complex reporting queries.
- Designed and implemented features for enterprise software products using **C#** and **.NET** frameworks.
- Collaborated with program managers to translate requirements into technical specifications and delivery plans.
- Participated in product testing and quality assurance processes, reducing post-release defect rates significantly.

Software Design Engineer Intern

Microsoft | 06/2006 - 08/2006

- Worked alongside full-time engineers on feature development and bug fixes for production systems.
- Contributed to internal tooling development, automating manual processes and improving team productivity.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007